James D. Motes
University of Illinois Urbana-Champaign
jmotes2@illinois.edu | jamesmotes.github.io

RESEARCH PROFILE

I am a robotics researcher developing planning systems that enable a multi-robot team to work together on complex tasks. My ongoing and future research builds on these foundations to help people direct robotic work and make decisions alongside a multi-robot team.

ACADEMIC APPOINTMENT

Postdoctoral Researcher, University of Illinois Urbana-Champaign August 2023–Present
Advisor: Dr. Nancy M. Amato

EDUCATION

Ph.D. in Computer Science, University of Illinois Urbana-Champaign August 2023
Ph.D. Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces
Advisor: Dr. Nancy M. Amato

M.S. in Computer Science, Texas A&M University August 2019
M.S. Thesis: Interaction Templates for Multi-Robot Systems
Advisor: Dr. Nancy M. Amato

B.S. in Computer Engineering, Texas A&M University May 2018
Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar

RESEARCH LEADERSHIP

- **Research direction.** Currently lead research teams involving 12+ graduate students in the Parasol Lab, setting project directions and coordinating research and publication development.
- **Graduate supervision and mentorship.** Second supervisor for a Ph.D. student at King's College London; mentored 20+ graduate students, including primary day-to-day mentorship of two Ph.D. and three M.S. graduates.
- **Open-source research software.** Primary developer and architectural lead of CoMotion; direct graduate-student development of Open-SPITE and coordinate development of the Parasol Planning Library.
- **Proposals and collaborations.** Primary author of an NSF robotics proposal (pending) and eight industry proposals; coordinate interdisciplinary and industry research collaborations.

PUBLICATIONS & SCHOLARLY OUTPUTS

Publication Overview

- 16 peer-reviewed archival publications.
- 4 first-author peer-reviewed publications and 2 first-author manuscripts under review.
- 11 student-led archival publications directly mentored (8 as a postdoctoral researcher), with additional manuscripts under review.
- Additional scholarly outputs include preprints, technical reports, workshop papers, and extended abstracts.
- Google Scholar: 429 citations, h-index: 7, as of October 15, 2026.

Notation

- An asterisk (*) indicates a student for whom Motes served as primary day-to-day research mentor.

Peer-Reviewed Archival Publications

As a Postdoctoral Researcher

[P16] **J.D. Motes**, M. Morales, and N. M. Amato, "P-ARC: Exploiting Subproblem Independence for Parallel Multi-Robot Motion Planning", **IEEE Robotics and Automation Letters (RA-L)**, 2026.

[P15] S. Lee*, **J.D. Motes**, I. Ngui*, M. Morales, and N. M. Amato, "Lazy-DaSH: A Lazy Approach for Hypergraph-Based Multirobot Task and Motion Planning", **IEEE Transactions on Robotics (T-RO)**, vol. 42, 2026.

[P14] C. McBeth*, **J.D. Motes**, I. Ngui*, M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning Using Workspace Guidance-Informed Hypergraphs", **IEEE Robotics and Automation Letters (RA-L)**, vol. 11, 2026.

[P13] H. Lee*, **J.D. Motes**, M. Morales, and N. M. Amato, "An Analysis of Constraint-Based Multiagent Pathfinding Algorithms", **IEEE Transactions on Robotics (T-RO)**, vol. 42, 2026.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.

[P12] I. Ngui*, C. McBeth*, A. Santos, G. He, K. J. Mimnaugh, **J.D. Motes**, L. Soares, M. Morales, and N. M. Amato, "ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality", in **International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)**, 2026.
- Presented at *ICAT-EGVE 2026*, Goiânia, Brazil, Sept. 2026.

[P11] I. Ngui*, C. McBeth*, **J.D. Motes**, M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement", in **Algorithmic Foundations of Robotics XVII: Proc. of the Seventeenth World Symposium on the Algorithmic Foundations of Robotics (WAFR)**, 2026.

[P10] S. Ashur*, M. Lusardi, M. Markowicz*, **J.D. Motes**, M. Morales, S. Har-Peled, and N. M. Amato, "SPITE: Simple Polyhedral Intersection Techniques for Motion Planning in Modified Environments", in **Algorithmic Foundations of Robotics XVI, Volume 1**, Springer Proceedings in Advanced Robotics, vol. 37, Springer, 2026.
- Presented at *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.

[P9] M. Qin*, I. Solis*, **J.D. Motes**, M. Morales, and N. M. Amato, "K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning", **IEEE Robotics and Automation Letters (RA-L)**, vol. 10, 2025.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, June 2026.

[P8] A. Goddu, T. Brown, M. Paton, **J.D. Motes**, and T. Chen, "Modifying ABIT* for Tethered Rappelling Robot Motion Planning", in **Proc. 2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)**, 2025.

[P7] I. Solis*, **J.D. Motes**, M. Qin*, M. Morales, and N. M. Amato, "Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-Robot Motion Planning", **IEEE Robotics and Automation Letters (RA-L)**, 2024.
- Presented at *IEEE ICRA@40*, Rotterdam, The Netherlands, Sept. 2024.

As a Graduate Student

[P6] C. McBeth*, **J.D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, "Scalable Multi-Robot Motion Planning for Congested Environments With Topological Guidance", **IEEE Robotics and Automation Letters (RA-L)**, 2023.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.

[P5] **J.D. Motes**, T. Chen, T. Bretl, M. Morales, and N. M. Amato, "Hypergraph-Based Multi-Robot Task and Motion Planning", **IEEE Transactions on Robotics (T-RO)**, 2023.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, May 2024.

[P4] H. Lee*, **J.D. Motes**, M. Morales, and N. M. Amato, "Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding", **IEEE Robotics and Automation Letters (RA-L)**, vol. 6, 2021.
- Presented at *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Prague, Czech Republic, 2021 (virtual).

[P3] I. Solis*, **J.D. Motes**, R. Sandström, and N. M. Amato, "Representation-Optimal Multi-Robot Motion Planning Using Conflict-Based Search", **IEEE Robotics and Automation Letters (RA-L)**, vol. 6, 2021.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Xi'an, China, 2021 (virtual).

[P2] **J.D. Motes**, R. Sandström, H. Lee*, S. Thomas, and N. M. Amato, "Multi-Robot Task and Motion Planning With Subtask Dependencies", **IEEE Robotics and Automation Letters (RA-L)**, vol. 5, 2020.
- Presented at *IEEE International Conference on Robotics and Automation (ICRA)*, Paris, France, 2020 (virtual).

[P1] **J.D. Motes**, R. Sandström, W. Adams, T. Ogunyale*, S. Thomas, and N. M. Amato, "Interaction Templates for Multi-Robot Systems", **IEEE Robotics and Automation Letters (RA-L)**, vol. 4, 2019.
- Presented at *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Macao, China, 2019.

Manuscripts Under Review

First Author

[U6] **J.D. Motes**, M. Morales, and N. M. Amato, "Multi-Robot Motions in Milliseconds: Vector-Accelerated Primitives for Sampling-Based Planning", *arXiv* preprint arXiv:2604.23960, 2026.

[U5] **J.D. Motes**, M. Morales, and N. M. Amato, "AO-ARC: Almost-Surely Asymptotically Optimal Multi-Robot Motion Planning with ARC", *arXiv* preprint arXiv:2606.27495, 2026.

Mentored Student-Led Manuscripts

[U4] V. Murta*, I. B. Love*, **J.D. Motes**, M. Morales, and N. M. Amato, "RoboCheck: A Fast, Exact Massively Parallel Collision Checker for Mobile Robotics", submitted, 2026.

[U3] M. Markowicz*, **J.D. Motes**, M. Morales, and N. M. Amato, "MR-SPITE: Accelerating Multi-Robot Conflict Scans via Hierarchical Swept-Volume Approximations", submitted, *arXiv* preprint arXiv:2609.23928, 2026.

[U2] Y. Arad*, M. Markowicz*, S. Ashur*, **J.D. Motes**, M. Morales, and N. M. Amato, "Serialized Red-Green-Gray: Quicker Heuristic Validation of Edges in Dynamic Roadmap Graphs", *arXiv* preprint arXiv:2603.28674, 2026.

[U1] I. Solis*, **J.D. Motes**, J. Montalvo, M. Qin*, M. Morales, and N. M. Amato, "Experience-Based Subproblem Planning for Multi-Robot Motion Planning", *arXiv* preprint arXiv:2411.08851, 2024.

Additional Scholarly Outputs

Preprints and Technical Reports

[TR2] H. Lee*, Z. Serlin, **J.D. Motes**, B. Long, M. Morales, and N. M. Amato, "PRISM: Complete Online Decentralized Multi-Agent Pathfinding With Rapid Information Sharing Using Motion Constraints", *arXiv* preprint arXiv:2505.08025, 2025.

[TR1] A. Attali, S. Ashur*, I. B. Love*, C. McBeth*, **J.D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, "A Framework for Guided Motion Planning", *arXiv* preprint arXiv:2404.03133, 2024.

Workshop Papers and Extended Abstracts

[W9] M. Markowicz*, S. Ashur*, **J.D. Motes**, and N. M. Amato, "Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps", in **4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments (ICRA 2025)**, Atlanta, GA, USA, May 19, 2025.

[W8] H. Lee*, **J.D. Motes**, Z. Serlin, M. Morales, and N. M. Amato, "Distributed Constraint-Based Search Using Multi-Hop Communication", in **ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)**, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W7] I. Ngui*, **J.D. Motes**, H. Lee*, M. Morales, and N. M. Amato, "Ergonomic Motion Planning for Human-Robot Collaboration", in **ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)**, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W6] K. Elimelech, **J.D. Motes**, M. Morales, N. M. Amato, M. Vardi, and L. E. Kavraki, "Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs", in **ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)**, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W5] S. Lee*, **J.D. Motes**, I. Ngui*, M. Morales, and N. M. Amato, "Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-Robot Task and Motion Planning", in **ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)**, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W4] M. Lusardi, M. Markowicz*, S. Ashur*, **J.D. Motes**, M. Morales, S. Har-Peled, and N. M. Amato, "SPITE: Simple Polyhedral Intersection Techniques for Modified Environments", in **ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)**, Rotterdam, The Netherlands, Sept. 23–26, 2024.

[W3] I. Ngui*, S. Lee*, **J.D. Motes**, M. Morales, and N. M. Amato, "A Hierarchical Approach to Workstation-based Task-Allocation and Motion Planning", in **RAFF2023: Robotics & AI in Future Factory (IROS 2023 Workshop)**, IROS 2023, Detroit, MI, USA, Oct. 1–5, 2023.

[W2] A. Attali, S. Ashur*, I. B. Love*, C. McBeth*, **J.D. Motes**, D. Uwacu, M. Morales, and N. M. Amato, "Evaluating Guiding Spaces for Motion Planning", in **Evaluating Motion Planning Performance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022)**, Kyoto, Japan, Oct. 23, 2022.

[W1] T. Chen, Z. Huang, **J.D. Motes**, J. Geng, Q. Ta, H. Dinkel, H. Abdul-Rashid, J. Myers, Y. Mun, W. Lin, Y. Huang, S. Liu, M. Morales, N. M. Amato, K. Driggs-Campbell, and T. Bretl, "Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration", in **IEEE ICRA Workshop on Collaborative Robots and the Work of the Future**, Philadelphia, USA, 2022.

OPEN-SOURCE RESEARCH SOFTWARE

CoMotion — Coordinated Motion Planning for Multi-Robot Systems
Research Software Lead
Open-source C++ platform integrating selective-coordination multi-robot planning with accelerated planning infrastructure, including ARC-family methods, parallel planning, and accelerated motion validation. Primary developer of the library's code; lead its architecture, technical direction, and ongoing development.

Open-SPITE
Research Software Lead
Open-source C++ library for efficient motion validation and replanning in changing environments, including SPITE and Serialized Red-Green-Gray (SerRGG). Designed the architecture and development plan; direct and coordinate graduate-student implementation, integration, and maintenance.

Parasol Planning Library (PPL)
Research Software Lead
Shared C++ robotics planning infrastructure supporting Parasol Lab research. Previously the library's leading developer; current responsibilities focus on managing the codebase and coordinating development.

PROPOSAL & SPONSORED RESEARCH DEVELOPMENT

National Science Foundation (NSF) — Advanced Manufacturing, Design, and Operations (AMDO) Proposal
October 2026 | Status: Pending
- Originated the concept and shaped the research questions and vision for robotic workcell design tools for non-expert users, building on the AR/VR platform [P12] and integrated planning systems.
- Drew on established industry connections to inform the research direction and secured a letter of collaboration.

National Science Foundation (NSF) – Engineering Research Center (ERC) Site Visit March 2026
- Supported preparation for an NSF Engineering Research Center site visit by developing research-thrust narrative and presentation materials.

National Science Foundation (NSF) — Foundational Research in Robotics (FRR) Proposal
September 2025 | Status: Pending
- Primary author of proposal to customize robotic motion behaviors for non-expert users through natural language preference specification.
- Coordinated collaboration with natural language processing researchers at the University of Illinois Urbana-Champaign.
- Developed research narrative connecting human preference specification, motion planning, and human-steerable robot behavior.

Industry Proposals and White Papers April 2025–Present
- Primary author of eight proposals and secondary author on four more.
- Proposal topics include human-robot interaction, multi-robot planning, environment design, and data-driven planning.
- Developed sponsored research narratives connecting scalable robot-team planning to deployable industrial automation systems.

INVITED TALKS & SEMINARS

Deciding What to Plan Together: DaSH for Multi-Robot Planning
- Robot Interactive Intelligence (RobIn) Lab, Department of Computer Science, The University of Texas at Austin, Austin, TX.
- October 2026. Invited lab seminar.

Adaptive Robot Coordination: Lessons, Limitations, and Opportunities
- Aggie Autonomous Robotics Research Group, Department of Computer Science and Engineering, Texas A&M University, College Station, TX.
- August 2026. Invited lab seminar.

Lessons from Manufacturing Startups
- AI Meets Robots & Machines Seminar, University of Illinois Urbana-Champaign, Urbana, IL.
- February 2026. Invited seminar.

Multi-Robot Task and Motion Planning in Hybrid State Spaces
- Kavraki Lab, Department of Computer Science, Rice University, Houston, TX.
- February 2024. Invited lab seminar.

Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules
- Center for Autonomy Portfolio Discovery Workshop, University of Illinois Urbana-Champaign, Urbana, IL.
- October 2019. Invited workshop talk.

TEACHING & CURRICULUM DEVELOPMENT

AI4ALL August 2021–December 2023
- Teaching assistant for Discover AI at the University of Illinois Urbana-Champaign (Fall 2021); instructor (Spring 2022); lead instructor (Fall 2022–December 2023).
- Instructor for introductory artificial intelligence courses delivered through partner programs at The University of Texas at El Paso, New Mexico State University, Worcester Polytechnic Institute, and Texas A&M University.
- Developed curriculum for the Discover AI introductory artificial intelligence course.
- Led and mentored instructor teams across participating universities.

Teaching Assistant, University of Illinois Urbana-Champaign January 2021–May 2021
- Teaching assistant for Computer Science Ph.D. seminar course.
- Organized guest speakers, managed grading, and coordinated student assignments.

Internal Lab Instruction, Parasol Lab August 2019–May 2021
- Developed and taught a motion-planning course for new Parasol Lab members.

MENTORING, RESEARCH SUPERVISION & OUTREACH

Mentoring has been central to my research since my second year of graduate study, with responsibilities growing from day-to-day project guidance to graduate-team leadership and formal doctoral supervision. I align projects with students' interests, providing close technical guidance initially and progressively transferring ownership of research directions and design decisions. This approach has helped me grow a collaborative research program through student-led work and develop researchers who now lead and mentor teams in academia and industry.

Second Supervisor, King's College London June 2026–Present
Andres Zamora, Ph.D. student.
Research: Learning by abstraction methods for multi-robot planning problems.
Primary supervisor: Dr. Khen Elimelech.

Graduate Research Leadership & Mentorship, Parasol Lab
- Mentored 20+ graduate students across doctoral and postdoctoral work; currently lead research teams involving 12+ graduate students.
- Set project directions and coordinate research milestones, paper development, and shared software integration. Mentorship contributed to 11 student-led archival publications.

Selected Completed Graduate Research Projects

Served as the primary day-to-day research mentor throughout the following projects.

- **Irving Solis, Ph.D. (2024)** — Multi-robot motion planning.
- **Hannah Lee, Ph.D. (2024)** — Multi-agent path finding; parallel and task-and-motion variants of conflict-based search.
- **Mike Qin, M.S. (2024)** — Multi-robot kinodynamic planning.
- **Yulie Arad, M.S. (2026)** — Dynamic-roadmap validation and replanning, GPU-accelerated planning.
- **Jared Buabeng, M.S. (2026)** — Natural language-conditioned motion planning.

Undergraduate Research Mentorship

- During graduate study, mentored undergraduate researchers, including local university students and DREU program participants.

Houston Robotics Club August 2023–Present

- Mentor community college and high school students on AI and robotics projects.
- Supported students who later enrolled in STEM undergraduate programs or obtained internships with local robotics companies.

PROFESSIONAL SERVICE

Report-Writing Team Lead, 2026 Computing Research Association Summit

- Led the organization of the report-writing team, contributed conference notes, compiled the report covering Summit talks and discussions, and authored the executive summary.

Program Committee Member

- World Symposium on Algorithmic Foundations of Robotics, 2026.
- World Symposium on Algorithmic Foundations of Robotics, 2024.

Reviewer/Referee August 2018–Present

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Symposium on Multi-robot and Multi-agent Systems (MRS)
- World Symposium (formerly International Workshop) on Algorithmic Foundations of Robotics (WAFR)
- International Journal of Robotics Research (IJRR)
- Artificial Intelligence Journal (AIJ)

Professional Memberships

- IEEE Robotics and Automation Society May 2019–Present

INDUSTRY, ENTREPRENEURSHIP & RESEARCH TRANSLATION

Optigon Inc., Principal Software Engineer, part-time May 2024–May 2026

- Led software development for DOE SBIR-funded high-throughput, non-contact spectroscopic metrology and analysis tools for clean-energy materials, including perovskite photovoltaics.
- Developed machine learning analytical models of solar cell behavior.

Normandy Automation, Founder May 2023–August 2025

- Developed autonomous robotic manufacturing concepts for just-in-time manufacturing with minimal human reprogramming.
- Leveraged AI and multi-robot systems for autonomous welding and CNC machine-tending workflows.
- Wrote NSF SBIR proposals for autonomous robotic manufacturing systems.

Industrial Human-Robot Factory Collaboration, UIUC August 2019–August 2023

- Collaborated with Foxconn Interconnect Technology and UIUC research groups led by Timothy Bretl and Katherine Driggs-Campbell on collaborative human-robot factory projects.
- Connected multi-robot task and motion planning research to industrial assembly and human-robot collaboration settings.

HONORS

Engineering Graduate Merit Fellowship August 2018–August 2019
President's Endowed Scholarship August 2014–May 2018
Distinguished Student Award – Dwight Look College of Engineering May 2017
Industrial Affiliates Program Scholarship August 2014–May 2018
Dell Merit Scholarship August 2017–May 2018